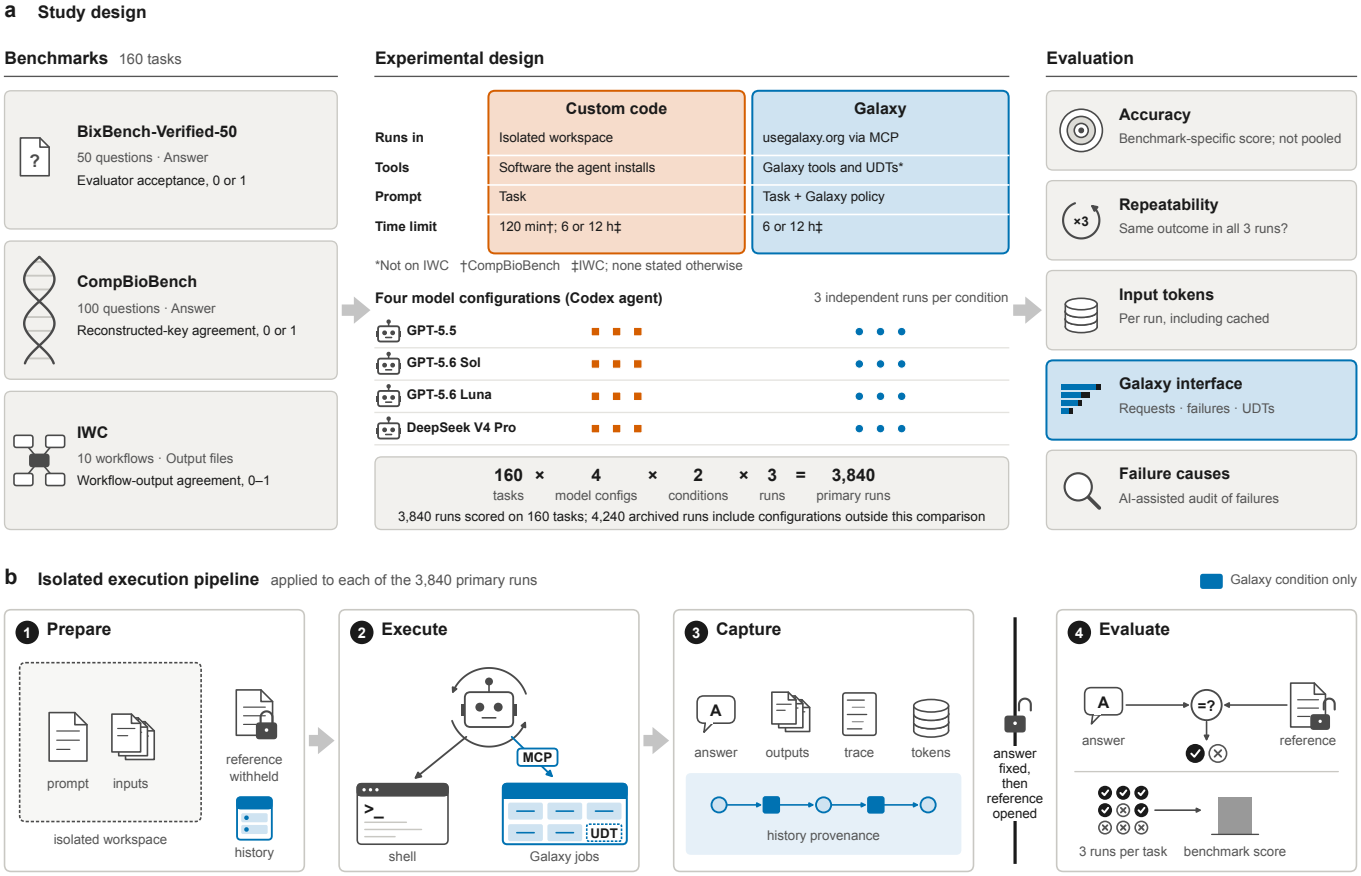

a Study design
Benchmarks 160 tasks
BixBench-Verified-50
50 questions · Answer
Evaluator acceptance, 0 or 1
CompBioBench
100 questions · Answer
Reconstructed-key agreement, 0 or 1
IWC
10 workflows · Output files
Workflow-output agreement, 0–1
Experimental design
Custom code
Galaxy
Runs in
Isolated workspace
usegalaxy.org via MCP
Tools
Software the agent installs
Galaxy tools and UDTs*
Prompt
Task
Task + Galaxy policy
Time limit
120 min†; 6 or 12 h‡
6 or 12 h‡
*Not on IWC †CompBioBench ‡IWC; none stated otherwise
Four model configurations (Codex agent)
3 independent runs per condition
GPT-5.5
GPT-5.6 Sol
GPT-5.6 Luna
DeepSeek V4 Pro
160 × 4 × 2 × 3 = 3,840
tasks
model configs
conditions
runs
primary runs
3,840 runs scored on 160 tasks; 4,240 archived runs include configurations outside this comparison
Evaluation
Accuracy
Benchmark-specific score; not pooled
Repeatability
Same outcome in all 3 runs?
Input tokens
Per run, including cached
Galaxy interface
Requests · failures · UDTs
Failure causes
AI-assisted audit of failures
b Isolated execution pipeline
applied to each of the 3,840 primary runs
Galaxy condition only
1 Prepare
prompt
inputs
isolated workspace
reference withheld
history
2 Execute
MCP
shell
Galaxy jobs
UDT
3 Capture
answer
outputs
trace
tokens
history provenance
answer fixed, then reference opened
4 Evaluate
answer
reference
3 runs per task
benchmark score